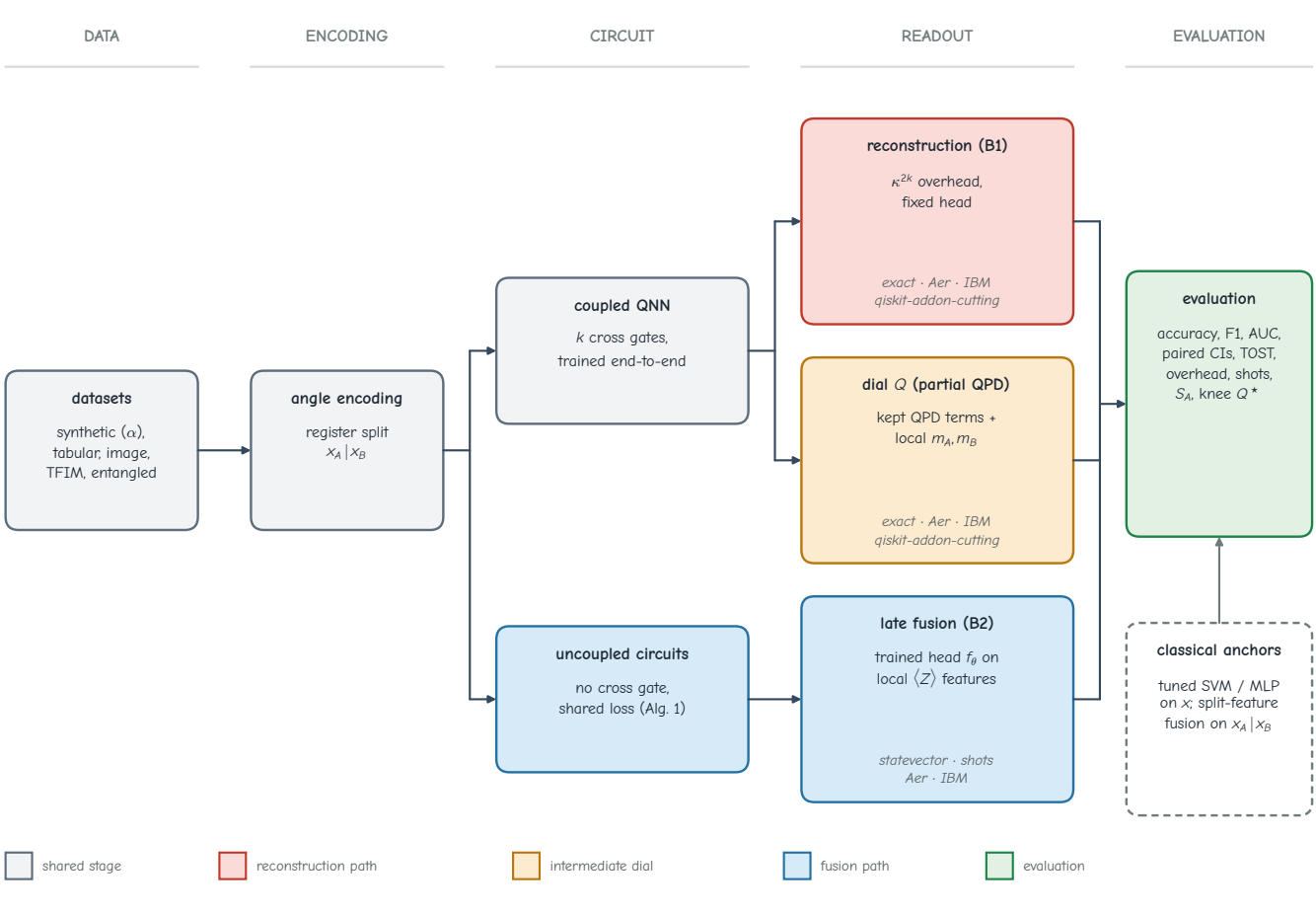
DATA
ENCODING
CIRCUIT
READOUT
EVALUATION
datasets
synthetic (α), tabular, image, TFIM, entangled
angle encoding
register split $x_A \mid x_B$
coupled QNN
k cross gates, trained end-to-end
uncoupled circuits
no cross gate, shared loss (Alg. 1)
reconstruction (B1)
κ^{2k} overhead, fixed head
exact · Aer · IBM qiskit-addon-cutting
dial Q (partial QPD)
kept QPD terms + local m_A, m_B
exact · Aer · IBM qiskit-addon-cutting
late fusion (B2)
trained head f_θ on local $\langle Z \rangle$ features
statevector · shots Aer · IBM
evaluation
accuracy, F1, AUC, paired CIs, TOST, overhead, shots, S_A, knee $Q^\star$
classical anchors
tuned SVM / MLP on x; split-feature fusion on $x_A \mid x_B$
shared stage
reconstruction path
intermediate dial
fusion path
evaluation